Week 1 CUSF Controls Freshers Project - Introduction

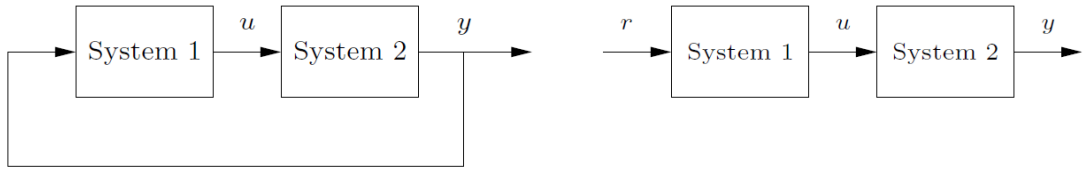
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Motivation

- Spacecrafts are highly complex systems
- Disturbances are everywhere and we can't predict everything that will happen
- We need a way to automatically correct the motion of the spacecraft against disturbances
- Lower level control systems will be present all throughout the system of the spacecraft
- Control systems use feedback

What is Feedback?



(a) Closed loop (b) Open loop

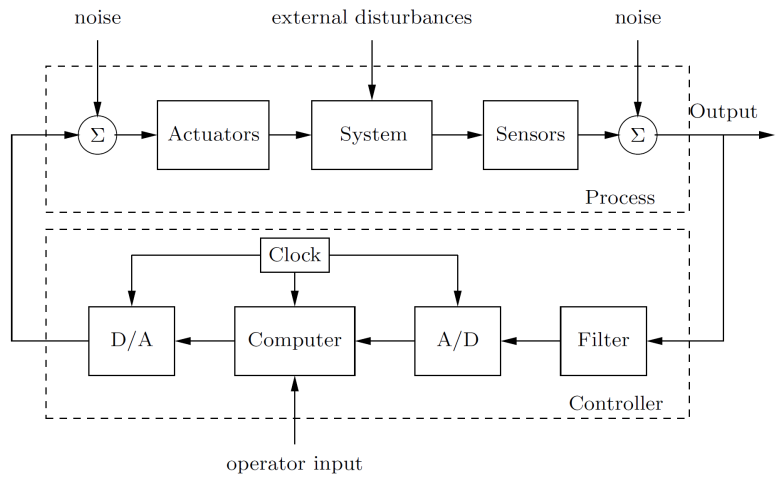
- Dynamical systems are systems whose behaviour change with time
- External forces or Disturbances
- Systems takes inputs and produces outputs
- Feedback is when the output of the system affects the input
- Control theory is a subject developed to formally approach the study of feedback
- Closed loop: Systems with feedback
- Open loop: Systems without feedback

Feedforward?

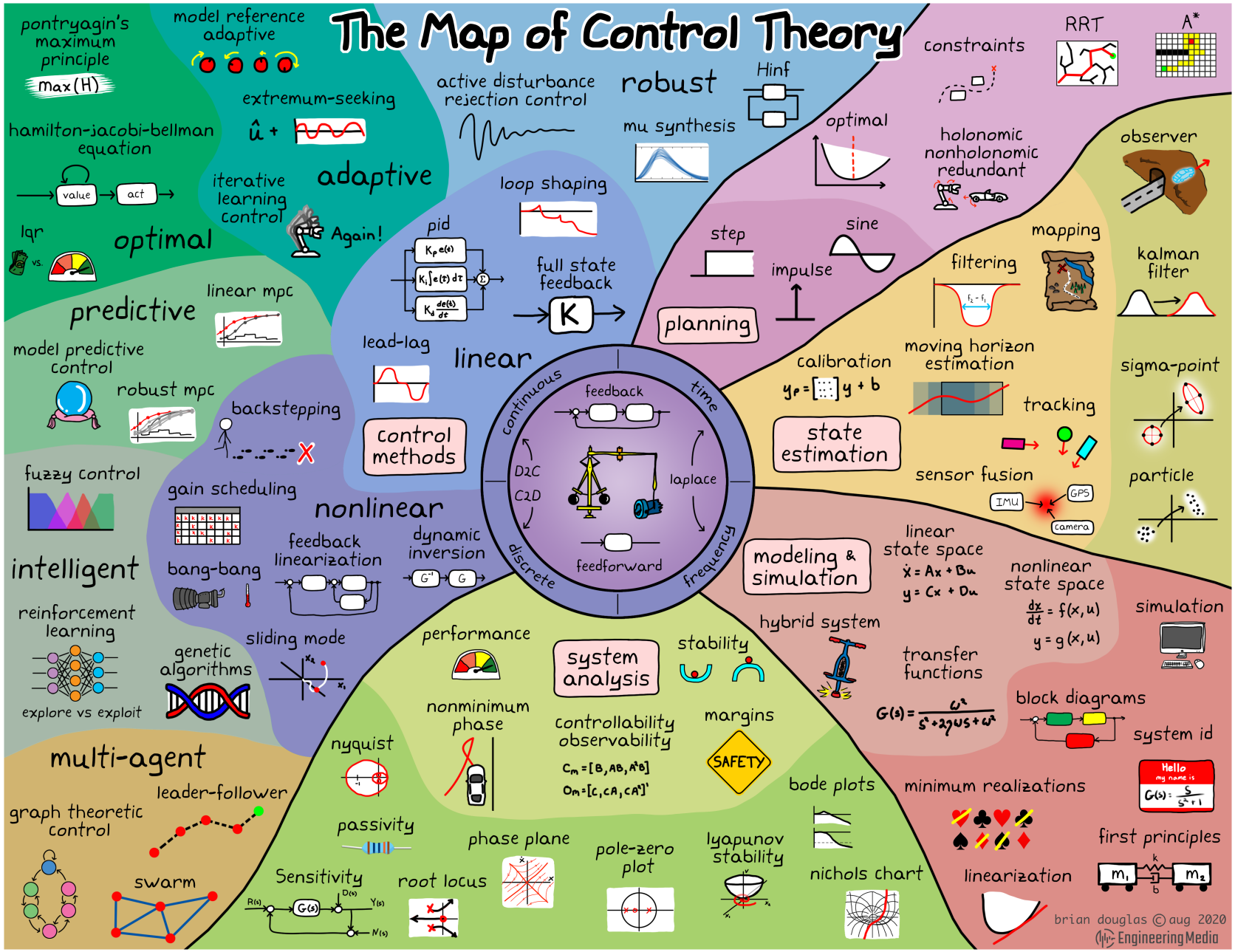
- Feedback is reactive
- Feedforward measures disturbances before it enters a system
- Informs corrective action before the disturbance affects the system

What is Control?

- Use of feedback and algorithms to change an engineering system to a desired state
- Performance
- Stability
- Guarantees
- Robustness to uncertainty and disturbances
- Sensors
- Implementation happens through software onto microcontrollers, with C or C++



Map of Control theory



What we are going to do

- System modelling - 3DOF model of a thrust vector controlled rocket
- Controller design - full state feedback, LQRs
- Observer design - linear observer design and Kalman Filters
- Simulations in MATLAB Simulink
- And more!

Further Resources: Links!

- Astrom and Murray's textbook: Feedback Systems PDF - Great introduction to linear control theory and frequency domain methods
- Brian Douglas's Youtube Channel: Brian Douglas
- Steve Brunton's Youtube Channel: Steve Brunton
- Journals: IFAC, IEEE, automatica
- Podcast: InControl
- Many, many, many textbooks